

Derived

The Calabi - Yau Auslander - Iyama Correspondence

(joint work with Fernando Muro)

Fix k : perfect field (e.g. $\text{char } k = 0$ or $k = \bar{k}$) & $d \geq 1$

Thm (A) (D) (Muro 2022 $d=1$, J-Muro 2022⁺ $d \geq 1$, J-Muro 2025⁺)

There is a 1:1 correspondence between the following:

(1) Quasi-iso classes of dg algebras A s.t.

(a) $H^0(A)$ is a basic fin.dim. alg.

(b) $A \in \text{perf}(A)$ is $d\mathbb{Z}$ -cluster tilting ($\Rightarrow A \cong A[d]$ in $\text{perf}(A)$)

(c) $A[n] \cong DA$ in $\mathcal{D}(A^e)$, $n \in d\mathbb{Z}$ A is bimodule right n -CY

(2) Equivalence classes of pairs (Λ, I) s.t. ↖ twisted $(d+2)$ -periodic

(a) Λ is a basic Frobenius alg.

(b) $I \in \text{mod}(\Lambda^e)$ $_{\Lambda}$ -invertible s.t. $\exists [\gamma] \in \text{Ext}_{\Lambda^e}^{d+2}(\Lambda, I)$

$$\gamma: 0 \rightarrow I \rightarrow \underbrace{P_{d+1} \rightarrow P_d \rightarrow \dots \rightarrow P_1 \rightarrow P_0}_{\cap \text{proj}(\Lambda^e)} \rightarrow \Lambda \rightarrow 0$$

$$\Lambda := \bigoplus_{d \in d\mathbb{Z}} \tau^d \Lambda, \text{ where } \tau \in \text{Aut}(\Lambda) \text{ satisfies } I \cong_{\Lambda} \tau I$$

(c) $\exists \varphi: \Lambda(n) \xrightarrow{\sim} D\Lambda$, $n \in d\mathbb{Z}$, in $\text{mod}^{\mathbb{Z}}(\Lambda^e)$ s.t.

↘ Universal Massey Product $\in \text{HH}^{d+2, -d}(\Lambda, \Lambda)$

$$\Delta_{\varphi}(\{m_{d+2}^{\mathbb{Z}}\}) = 0 \text{ in } \text{HH}^{d+1, -d}(\Lambda, \Lambda)$$

↗ Batalin - Vilkoviski operator

The correspondence (1) $\rightarrow$ (2) is given by $A \mapsto (H^0(A), H^{-d}(A))$

Moreover, for A as in (1), $\text{perf}(A)$ admits a unique dg enhancement.

Def (Iyama-Yoshino 2008, Geiss-Keller-Oppermann 2013)

$H^*(A)$
degree wise
fin. dim. $A \in \text{perf}(A)$: basic ^{fixed} is $d\mathbb{Z}$ -cluster-tilting if $A \cong A[d]$ and

$$\text{add}(A) = \{ X \in \text{perf}(A) \mid \forall 0 < i < d \text{ Hom}_A(A, X[i]) = 0 \}$$

$$= \{ X \in \text{perf}(A) \mid \forall 0 < i < d \text{ Hom}_A(X, A[i]) = 0 \}$$

Remark $A \in \text{perf}(A)$ is $1\mathbb{Z}$ -CT $\Leftrightarrow \text{add}(A) = \text{perf}(A)$

Remark $A \in \text{perf}(A)$ is $d\mathbb{Z}$ -CT $\Rightarrow H^*(A) = H^{d\mathbb{Z}}(A)$
 $A[n] \cong DA$ in $\mathcal{D}(A^e) \Rightarrow H^*(A)(n) \cong DH^*(A)$ as ^{graded} $H^*(A)$ -bimodules

Thm (Dugas 2012, Green-Susashall-Solberg 2003 + GKO 2013
 + Amiot 2007 $d=1$ / Liu 2019 $d \geq 1$ + Hanikava 2020)

$A \in \text{perf}(A)$: basic $d\mathbb{Z}$ -CT $\Rightarrow (H^0(A), H^{-d}(A))$ satisfies conditions (2a) & (2b) in Thm A.

Remark $A \in \text{perf}(A)$: $d\mathbb{Z}$ -CT $[d] \subset \text{add}(A) \xrightarrow{\sim} \text{proj}(H^0(A)) \hookrightarrow - \otimes_{H^0(A)} H^d(A)$

Example $A = k\langle z^{\pm} \rangle$, $|z| = -d$. Easy: $A \in \text{perf}(A)$ is $d\mathbb{Z}$ -CT, $H^0(A) = k$
 $A[d] \cong DA$ in $\mathcal{D}(A^e)$

Example $\Pi := (d+1)$ -CY completion of a d -rep.-finite d -hereditary alg.
 $A := R\text{End}(\Pi)$ in $\text{cosing}(\Pi) := \text{perf } \Pi / \text{pvd } \Pi$
 $A \in \text{perf}(A)$: $d\mathbb{Z}$ -CT and $H^0(A) \cong H^0(\Pi)$: $(d+1)$ -preproj. alg.
 $A[d] \cong DA$ in $\mathcal{D}(A^e)$

($d=1$) $\Pi :=$ derived 2-preprojective alg. of ADE quiver $H^0(\Pi)$: preproj. alg.
 $k = \mathbb{C}$ $\text{perf}(A) = \underline{\text{CM}}$ (corresponding ADE surface sing.)

($d=2$) $\Pi :=$ Ginzburg 3-CY dg alg of a self-injective QP. $H^0(\Pi)$: Jacobian algebra
 $\text{perf}(A)$ corresponding 2-CY Amiot cluster category.

A : dg algebra with the following properties

- $H^*(A) = H^{d\mathbb{Z}}(A)$, i.e. $H^i(A) \neq 0 \Rightarrow i \in d\mathbb{Z}$
- A is locally proper, i.e. $\forall i \in \mathbb{Z} \ H^i(A)$ is finite-dimensional
- $H^0(A)$ is a basic fin. dim. alg.
- $\exists \varphi \in H^d(A) \cong \text{Hom}_{\mathcal{D}(A)}(A, A[d])$ invertible

e.g. $A \in \text{perf}(A)$ is basic $d\mathbb{Z}$ -CT.

$$\Lambda := H^0(A), \quad \Lambda = \bigoplus_{\sigma} \langle \iota^{\pm} \rangle * H^0(A) \cong H^*(A), \quad |\iota| = -d, \quad \sigma: x \mapsto \varphi^{-1} x \varphi$$

Remark $\Lambda^{-d} \cong H^{-d}(A)$ as Λ -bimodules

Choose: minimal A_{∞} -algebra model $(H^*(A), m^A)$ for A .

Notice: $m_{i+2}^A \neq 0 \Rightarrow i \in d\mathbb{Z}$ since $H^*(A) = H^{d\mathbb{Z}}(A)$ & $|m_{i+2}^A| = -i$

$$\rightsquigarrow (H^*(A), m^A) = (\Lambda, m_{d+2}^A, m_{2d+2}^A, m_{3d+2}^A, \dots)$$

$m_{d+2}^A \in C^{d+2, -d}(\Lambda, \Lambda)$ is a Hochschild cocycle

$$\rightsquigarrow \{m_{d+2}^A\} \in HH^{d+2, -d}(\Lambda, \Lambda) : \text{Univ. Massey Product of length } d+2$$

$j: \Lambda \hookrightarrow \Lambda$ induces a map

$$j^*: HH^{d+2, -d}(\Lambda, \Lambda) \longrightarrow HH^{d+2, -d}(\Lambda, \Lambda)$$

$\psi \qquad \qquad \qquad \text{II2}$

$$\{m_{d+2}^A\} \longmapsto j^* \{m_{d+2}^A\} \in \text{Ext}_{\Lambda^e}^{d+2}(\Lambda, {}_1\Lambda_r)$$

$\uparrow$ restricted UMP of length $d+2$

Thm (J-Muro 2022⁺) For A as above, TFAE:

(1) $A \in \text{perf}(A)$ is (basic) $d\mathbb{Z}$ -CT

(2) $j^* \{m_{d+2}^A\} \in \text{Ext}_{\Lambda}^{d+2}(\Lambda, {}_1\Lambda_r)$ exhibits an isomorphism

$$j^* \{m_{d+2}^A\} : \Omega_{\Lambda^e}^{d+2}(\Lambda) \xrightarrow{\sim} {}_1\Lambda_r \quad \text{in } \underline{\text{mod}} \Lambda^e$$

Slogan The restricted UMP of length $d+2$ witnesses the $d\mathbb{Z}$ -CT property of the dg algebra A .

Thm (B) (J-Muro 2022⁺) A : dg alg. s.t. $H^*(A) = H^{d\mathbb{Z}}(A)$ and

$$HM^{p+2, -p}(H^*(A), \{m_{d+2}^A\}) = 0 \quad \text{for } p > d$$

$\uparrow$ Hochschild-Massey cohomology

If B is a dg algebra with $H^*(B) \cong H^*(A)$ and

$$\{m_{d+2}^B\} = \{m_{d+2}^A\} \in HH^{d+2, -d}(H^*(A), H^*(A))$$

then B is quasi-isomorphic to A .

Remark For $d=1$ and $HH^{3, -1}(H^*(A), H^*(A)) = 0$, Thm B recovers Kadeishvili's Intrinsic Formality Theorem.

$HM(H^*(A), \{m_{d+2}^A\})$ is the cohomology of

$\hookrightarrow$ Hochschild-Massey complex

$$CM^{p, q}(H^*(A), \{m_{d+2}^A\}) := HH^{*, *}(H^*(A), H^*(A)) \quad p \geq 2, q \in \mathbb{Z}$$

$$d_{HM} : x \longmapsto [\{m_{d+2}^A\}, x] \quad \text{of bidegree } (d+1, -d) \quad (\text{if char } k \neq 2)$$

Rule $d_{HM} \circ d_{HM} = 0$ uses $Sg(\{m_{d+2}^A\}) = 0$

Rule $CM(H^*(A), \{m_{d+2}^A\})$ is the $(d+1)$ -st page of an extended Bousfield-Kan spectral sequence (Muro 2020 $d=1$)

Aim Analogues of $\{m_{d+2}^A\}$ and Thm B for bimodules?

Λ : graded algebra Fix: M a graded Λ -bimodule

$$C_{\Lambda^e}^{n,r}(M, M) := \bigoplus_{p+q=n} C_{\Lambda^e}^{p,q,r}(M, M) \quad n \geq 0, \quad r \in \mathbb{Z}, \text{ where}$$

$$C_{\Lambda^e}^{p,q,r}(M, M) := \text{Hom}_{\Lambda^e}(\Lambda^{\otimes p} \otimes M \otimes \Lambda^{\otimes q}, M(r))$$

$$\rightsquigarrow H^{*,*}(C_{\Lambda^e}(M, M), -d_{\text{Bim}}) \cong \text{Ext}_{\Lambda^e}^{*,*}(M, M)$$

$C_{\Lambda^e}(M, M)$ is naturally a $C(\Lambda, \Lambda)$ -bimodule

We introduce the bimodule Hochschild cochain complex

$$C(\Lambda | M) := \text{cocone} \left(C(\Lambda, \Lambda) \xrightarrow{\quad \delta \quad} C_{\Lambda^e}(M, M) \right)$$

$\cup_{d_{\text{Hoch}}} \quad \quad \quad \cup_{-d_{\text{Bim}}}$

$$C^{n,r}(\Lambda | M) := C^{n,r}(\Lambda, \Lambda) \oplus C_{\Lambda^e}^{n-1,r}(M, M)$$

It has the bidegree $(1,0)$ differential

$$d_{\Lambda | M} := \begin{pmatrix} d_{\text{Hoch}} & 0 \\ \delta & d_{\text{Bim}} \end{pmatrix} \quad \delta : c \longmapsto s(\text{id}_M \cdot c - (-1)^{|c|} c \cdot \text{id}_M)$$

$HH^{*,*}(\Lambda | M) := H^{*,*}(C(\Lambda | M))$ is also a Gerstenhaber alg.

↑ bimodule Hochschild cohomology

By construction, there is a triangle of bigraded vect. spaces

$$C_{\Lambda^e}(M, M)[-1] \rightarrow C(\Lambda | M) \rightarrow C(\Lambda, \Lambda) \xrightarrow{\delta} C_{\Lambda^e}(M, M)$$

Remark $C(\Lambda, \Lambda)$ is to the endomorphism operad $\text{End}(\Lambda)$
 what $C(\Lambda | M)$ is to the linear end. operad $\text{End}(\Lambda, M)$

$$A: \text{dg. alg. st. } \Lambda := H^*(A) = H^{d\mathbb{Z}}(A)$$

Consider now M a dg A -bimodule s.t. $H^*(M) = H^{d\mathbb{Z}}(M)$
 (e.g. $M = DA$ or $M = A[n]$ with $n \in d\mathbb{Z}$)

Choose a minimal A^∞ -algebra model $(H^*(A), m^A)$ of A
 and a compatible minimal A^∞ -bimodule model

$$(H^*(M), m^M) \text{ of } M, \quad m_n^M = (m_{p,q}^M)_{p+q=n} \quad |m_n^M| = 2-n$$

$$m_{p,q}^M : \underbrace{H^*(A) \otimes \dots \otimes H^*(A)}_p \otimes \underbrace{H^*(M)}_1 \otimes \underbrace{H^*(A) \otimes \dots \otimes H^*(A)}_q \rightarrow H^*(M)$$

$$m_n^{A \ltimes M} := m_n^A + m_n^M \in C^{n, 2-n}(H^*(A) | H^*(M))$$

As before, $m_{i+2}^M \neq 0 \Rightarrow i \in d\mathbb{Z}$ since $H^*(M) = H^{d\mathbb{Z}}(M)$

$m_{d+2}^{A \ltimes M} \in C^{d+2, -d}(H^*(A) | H^*(M))$ is a cycle

$\rightsquigarrow \{m_{d+2}^{A \ltimes M}\} \in HH^{d+2, -d}(H^*(A) | H^*(M)), \quad Sq(\{m_{d+2}^{A \ltimes M}\}) = 0$
 $\uparrow$ bimodule UMP of length $d+2$

$CM_{\Lambda^e}^{n,r}(H^*(M), \{m_{d+2}^{A \ltimes M}\}) := \text{Ext}_{\Lambda^e}^{n,r}(H^*(M), H^*(M)) \quad n \geq 1, r \in \mathbb{Z}$
 $\uparrow$ bimodule Massey complex

with the bidegree $(d+1, -1)$ differential

↪ Bracket of $HH^{*,*}(H^*(A) | H^*(M))$

$$d_{\text{Bim HM}} : x \longmapsto [\{\omega_{d+2}^{A \times M}\}, x]$$

$$EM_{\wedge^e}^{*,*}(H^*(M), \{\omega_{d+2}^{A \times M}\}) := H^{*,*}(CM_{\wedge^e}(M, \{\omega_{d+2}^{A \times M}\}))$$

↑ bimodule Massey cohomology

Thm (B) for bimodules (J-Mun 2025⁺)

A : dg alg. s.t. $H^*(A) = H^{d\mathbb{Z}}(A)$ such that
 M : dg A -bim. s.t. $H^*(M) = H^{d\mathbb{Z}}(M)$

$$EM_{\wedge^e}^{p+1, -p}(H^*(M), \{\omega_{d+2}^{A \times M}\}) = 0 \quad \forall p > d$$

If N is a dg A -bim s.t. $H^*(N) \cong H^*(M)$ and

$$\{\omega_{d+2}^M - \omega_{d+2}^N\} = 0 \in \text{Ext}_{\wedge^e}^{d+1, -d}(H^*(M), H^*(M))$$

then N is quasi-isomorphic to M .

Prop (J-Mun 2025⁺) A : dg alg. s.t. $H^*(A) = H^{d\mathbb{Z}}(A)$

Suppose $\exists \varphi : H^*(A)(n) \xrightarrow{\sim} DH^*(A)$ for some $n \in \mathbb{Z}$

Then, TFAE :

↪ diagonal A -bimodule

$$(1) \quad \{\omega_{d+2}^{A[0]} - \omega_{d+2}^{DA[-n]}\} = 0 \in \text{Ext}_{\wedge^e}^{d+1, -d}(H^*(A), H^*(A))$$

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$$(2) \quad \Delta_{\varphi}(\{\omega_{d+2}^A\}) = 0 \in HH^{d+1, -d}(H^*(A), H^*(A))$$

↑ algebra UMP of length $d+2$

Thm (J-Muro 2025⁺) Λ : graded algebra

$$HH^{*,*}(\Lambda | \Lambda) \xrightarrow[\text{iso}]{\exists K} HH^{*,*}(\Lambda)[\varepsilon] \quad \varepsilon^2 = 0 \quad |\varepsilon| = (1, 0)$$

If $\Lambda = H^*(A) = H^{\text{dZ}}(A)$ for a dg algebra A , then

$$K: \{u_{d+2}^{A[0]} - u_{d+2}^{DA[n]}\} \longmapsto \Delta_\varphi(\{u_{d+2}^A\}) \cdot \varepsilon$$

Remark

$$\begin{array}{ccc}
 C(\Lambda | \Lambda) & \xrightarrow{\Psi} & C(\Lambda | \Lambda(n)) \\
 \gamma \downarrow & = & \downarrow \theta \\
 C(\Lambda | \Lambda) & \xleftarrow{\Phi} & C(\Lambda | D(\Lambda(n)))
 \end{array}$$

We identify $u^{A \ltimes DA[n]} = \gamma(u^{A \ltimes A[0]})$

$\uparrow$ induced by iso $\varphi: \Lambda(n) \xrightarrow{\sim} D\Lambda$

$$\Psi(u^{A \ltimes A[0]}) = u^{A \ltimes A[n]}, \quad \theta(u^{A \ltimes A[n]}) = u^{A \ltimes D(\Lambda[n])}$$

The Batalin-Vilkoviski operator (assoc. to $\varphi: \Lambda(n) \xrightarrow{\sim} D\Lambda$)

$\Delta = \Delta_\varphi: C^{m,r}(\Lambda | \Lambda) \longrightarrow C^{m-1,r}(\Lambda, \Lambda)$ determined by

$$\langle S^n \Delta(c)(x_1, \dots, x_{m-1}), x_m \rangle$$

$$= \sum_{i=1}^m (-1)^{i(m-1)+ks} \langle S^n c(x_i, \dots, x_m, x_1, \dots, x_{i-1}), 1_A \rangle_\varphi$$

$$\langle S^n x, y \rangle_\varphi := \varphi(S^n x)(y) \quad x, y \in \Lambda$$